Baichuan Li

Curriculum Vitae

Dallas, TX — bauchuanli2000@gmail.com — (217) 607-4443
https://baichuan0113.github.io

EDUCATION

Doctor of Philosophy in Operations Research
Years: 2025 - Present
Institution: Southern Methodist University
Department: Department of Operations Research and Engineering Management
Accomplished coursework: Linear Programming, Stochastic Modeling, Markov Decision Process, Network Flow.

Master of Science in Computer Science
Year of completion: 2025
Institution: Texas A&M University
Department: Department of Computer Science & Engineering
Accomplished coursework: Data structures, Algorithms Analysis, Data Science, Object Oriented Programming, Software Engineering, Artificial Intelligence, Computer Organizations, Virtual Reality, Data Visualization, Robotics & Spatial Intelligence, Human-AI Interaction.

Bachelor of Science in Mathematics
Year of completion: 2022
Institution: University of Illinois at Urbana-Champaign
Department:College of Liberal Arts & Sciences
Accomplished coursework: Linear Algebra, Probability Theory, Discrete Mathematics, Abstract Algebra, Calculus, Real Analysis.

EMPLOYMENT

Graduate Teaching Assistant
08/2025-Present
Institution: Southern Methodist University
Department: Department of Operations Research and Engineering Management
- OREM 3362 Production and Operations Engineering
- OREM 5357 Analytics for Decision Support

Graduate Research Assistant
08/2025-Present
Institution: Southern Methodist University
- Research on Bayesian Inference, Markov Decision Process and Large Language Model

Information Technology Intern
06/2023-08/2023
Institution: YAPP USA Automotive Systems, Inc.
Department: Technology department

• Offered hands-on technical support to end-users, skillfully resolving software-related issues, both onsite and remotely, leading to a 20% decrease in the overall number of support requests.

• Delivered renovate on company's official website, include updating the website content, adding product descriptions, images, technical specifications, and relevant documentation by CSS, HTML and JavaScript to ensure accuracy and consistency.

PUBLICATIONS

- **Baichuan Li**, Larry Powell, Tracy Hammond. "It's Not Just Labeling: A Research on LLM-Generated Feedback Interpretability and Image Labeling Sketch Features." *arXiv preprint arXiv:2505.19419*, 2025.
- Z. Zheng, **Baichuan Li**, J. Yao, L. Long. "Reliable Financial Named Entity Recognition under Domain Shift." *arXiv preprint arXiv:2608.19558*, 2026.
- **Baichuan Li**, J. Yao, Z. Zheng. "Remember, Verify, or Ask? Cross-Family Evaluation of Memory Commitment in LLM Agents." *arXiv preprint arXiv:2608.19564*, 2026.
- J. Yao, Z. Zheng, **Baichuan Li**. "Measuring Whether LLM Tutors Teach or Solve: A Diagnostic for Educational Impact." *arXiv preprint arXiv:2606.16206*, 2026.

• X. Sun, **Baichuan Li**, H. Pang. "Portfolio Construction for Pharmaceutical Industry." *E3S Web of Conferences*, vol. 275, 03032, 2021.

RESEARCH EXPERIENCE

Bayesian Information Seeking for LLM Agents (Southern Methodist University, manuscript submit to ICLR 2027)

• Developed a method for improving how large language models gather information and make decisions in multi-turn interactive settings, combining Bayesian inference with LLM fine-tuning.

• Built the full experimental pipeline in Python/PyTorch: data generation, model training on GPU clusters (Slurm), and a reproducible evaluation framework with statistical testing.

• Resulting model significantly outperforms the base model and strong reported baselines on a public multi-turn benchmark.

Research on LLM generated feedback interpretability and image labeling sketch features (MS Thesis Project)
05/2024-05/2025 **(Passed Thesis Defense on 03/05/2025)**
Institution: Sketch Recognition Lab, Texas A&M University

• Generated synthetic stroke data by extracting contours and performing stochastic resampling to mimic human free-hand sketches for image labeling tasks.

• Quantified LLM-generated feedback accuracy by using the RAGAS evaluation framework

• Compared various prompting strategies to understand how they influence LLM-generated feedback by non-parametric tests.

• Performed correlation analyses between sketch features and LLM feedback interpretability metrics, revealing modest correlations.

TECHNICAL SKILLS

•Programming Languages: Java, Python, C++, C, Ruby&Rails, MySQL, Shell Script, JavaScript, HTML, CSS.
•Frameworks: Express, Node, Pandas, OpenCV, React, Unity, Robot Operating System (ROS).
•Tech-Stack: Git, Cucumber, BootStrap, Linux, Github, Restful API.